

AgentGER: Toward a Human-Aligned Generation–Evaluation–Refinement Paradigm for Figure Understanding

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Abstract

Figure-to-text is a key task for assessing figure understanding capabilities of models. Existing approaches face two main challenges: 1) the high cost of constructing high-quality data and 2) the lack of fine-grained and interpretable evaluation aligned with human experts. Accordingly, we propose AgentGER, an agent framework that integrates capabilities of generation, evaluation, and refinement for figure-to-text. AgentGER includes a Generation Model that generates summaries with hierarchical quality levels, an Evaluation Model that performs multi-dimensional human-aligned assessment with a Chain-of-Evaluation mechanism, and a Refinement Model that improves summaries based on evaluation feedback. We further construct a large-scale dataset, named FigGER, with 11,000 summaries and 55,000 multi-dimensional scoring labels through a human–machine collaborative annotation. Experiments show that AgentGER significantly outperforms all baselines, even surpasses the strong Gemini-3.0-Pro, and achieves performance comparable to human experts on both evaluation and refinement tasks.

1 Introduction

Figures are compact and structured media widely used in scientific papers, technical reports, and various professional domains. (Masry et al., 2022, 2024). With the advancement of multimodal large language models (MLLMs) (Zhang et al., 2024a), figure understanding has become a crucial research domain bridging visual perception and textual reasoning (Krishna et al., 2017; Wang et al., 2025). Figure-to-text tasks represent one of the primary benchmarks for evaluating an MLLM’s figure understanding capabilities (Davila et al., 2021; Hsu et al., 2021; Lin et al., 2014; Xia et al., 2024).

In recent years, Figure-to-Text research has made significant progress. For data construction, various figure-related datasets have been

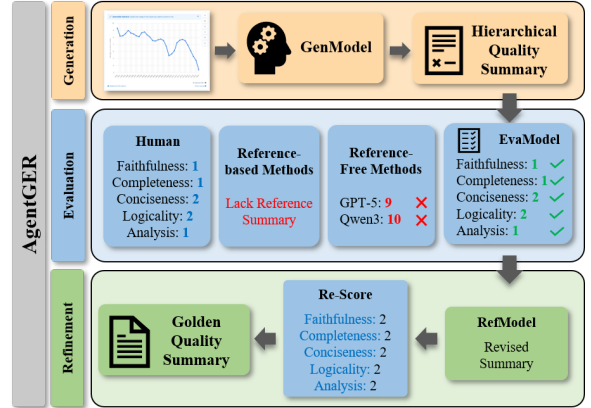


Figure 1: AgentGER consists of 3 models (GenModel, EvaModel, and RefModel) for generation, evaluation, and refinement.

built to support model training (Masry et al., 2022; Han et al., 2023a; Wang et al., 2025), including datasets from real-world figures (Kantharaj et al., 2022; Akhtar et al., 2024) and datasets using titles or descriptions as weak supervision (Rahman et al., 2023; Xia et al., 2024). For evaluation, existing methods mainly fall into two groups. Reference-based metrics, such as ROUGE (Lin, 2004), BLEU (Papineni et al., 2002), and BERTScore (Zhang et al., 2019), compare generated summaries with human-written references. Reference-free methods, including LLM-as-a-Judge, directly assess image-text alignment without reference summaries (Zhang et al., 2024b; Xia et al., 2024).

However, existing studies still face two key challenges: 1) High-quality training data is costly to obtain, making it difficult to balance scale and quality. Figure summarization requires domain expert knowledge and data analysis skills to capture trends, comparisons, and key findings. Thus, existing datasets rely on titles, captions, shallow descriptions, or templates, which limits diversity and analytical depth (Kantharaj et al., 2022; Han et al.,

2023b; Xia et al., 2024). 2) Current evaluation methods lack fine-grained and interpretable assessment aligned with human experts. Reference-based metrics focus on shallow semantic similarity, failing to determine whether summaries truly reflect figure information (Gigant et al., 2024). Reference-free methods provide a single overall score, lacking clear distinction across quality dimensions (Irugalbandara et al., 2024). This confuses the cause of model errors and limits direct guidance for refining summaries. These issues constrain target-guided refinement and reliable evaluation of figure-to-text models, as shown in Figure 1.

To address these challenges, we propose AgentGER, an agent framework for figure summarization and evaluation. AgentGER contains three models: Generation Model (GenModel), Evaluation Model (EvaModel), and Refinement Model (RefModel). GenModel takes figures as input and generates summaries at three quality levels: *low*, *medium*, and *high*. It is model-agnostic and can use different MLLMs to increase the diversity of candidate summaries. EvaModel evaluates summaries from five human-aligned dimensions (Yue et al., 2025): *Faithfulness*, *Completeness*, *Conciseness*, *Logicality*, and *Analysis*. For each dimension, it outputs a discrete score from $\{0,1,2\}$. Also, EvaModel introduces a *Chain-of-Evaluation (CoE)* mechanism, which generates dimension-wise reasoning before assigning scores. CoE makes the evaluation process more interpretable and provides explicit feedback for identifying weak dimensions in summaries. Following a curriculum learning strategy, AgentGER first trains EvaModel to learn reliable scoring and CoE-based assessment, and then trains RefModel to perform evaluation-guided refinement based on these feedback signals. RefModel uses the multi-dimensional scores and CoE rationales to improve weak dimensions while preserving correct content, forming an automated generation-evaluation-refinement paradigm.

To construct scalable data and expert-aligned evaluation methods, we design a human-machine collaborative paradigm. GenModel first generates 1,000 summaries with controlled quality levels. Human annotators score each summary across five dimensions and refine them into 1,000 high-quality gold summaries. Based on these seed data, we train EvaModel and RefModel, which are then used to generate 10,000 synthetic samples. From synthetic samples, 1,000 samples are selected through stratified sampling and re-annotated by experts to assess

quality and build a gold benchmark for figure-to-text evaluation. Extensive experiments show that AgentGER outperforms all baselines on the benchmark evaluation task, surpasses the strong Gemini-3.0-Pro, and achieves human-level performance on both evaluation and refinement tasks.

The main contributions are summarized as follows¹: 1) We propose AgentGER, an agent framework for figure-to-text tasks, enabling hierarchical quality summary generation, human-aligned interpretable evaluation, and target-guided summary refinement. 2) We construct a large-scale, high-quality figure-to-text dataset (FigGER) comprising 11,000 figure images, 11,000 summary texts, and 55,000 five-dimensional scoring labels with reasoning chains. 2,000 summaries and 10,000 scoring labels with reasoning chains are annotated by human experts, providing a reliable foundation for model training and evaluation. 3) Experimental results demonstrate that AgentGER significantly outperforms strong baseline models, achieving state-of-the-art performance. Also, AgentGER achieves capabilities approaching human experts in both figure-to-text evaluation and refinement task, showing the effectiveness of the proposed paradigm.

2 Related Works

2.1 Figure Understanding Data

Recent research on figure understanding mainly focuses on tasks such as figure question answering (Chaudhry et al., 2020; Singh and Shekhar, 2020; Methani et al., 2020), figure parsing (Meng et al., 2024; Xia et al., 2024), and figure-to-text generation (Akhtar et al., 2024). Early methods mainly relied on rules or templates to convert numerical values into fixed textual descriptions (Kahou et al., 2017; Kafle et al., 2018). With the development of MLLMs, recent approaches adopt end-to-end frameworks that jointly model visual structure, numerical information, and text (Masry et al., 2022; Xia et al., 2024). Correspondingly, several datasets based on real or synthetic figures have been proposed to support figure understanding and reasoning (Masry et al., 2023; Xu et al., 2023; Xia et al., 2023; Liu et al., 2024). However, due to the high cost of expert annotation, most existing datasets rely on templates or general MLLMs (Anthropic, 2025) for summary generation. Such limitations make models prone to generating fluent but factually incorrect or incomplete summaries.

¹All the code and dataset will be released upon acceptance.

2.2 Figure-to-Text Evaluation

Existing figure-to-text evaluation methods can be divided into reference-based and reference-free approaches. Reference-based methods, such as BLEU, ROUGE, BERTScore, and CIDEr (Vedantam et al., 2015), evaluate generated summaries by comparing them with reference texts. These methods are simple to implement and highly reproducible, making them widely adopted for evaluation. However, reference-based methods heavily rely on high-quality reference summaries (Gigant et al., 2024), which are costly to obtain. Additionally, these methods focus on surface-level semantic matching, limiting their ability to assess faithfulness and logicity to figure information (Zhang et al., 2019; Fabbri et al., 2021). In contrast, reference-free methods do not rely on reference summaries. They evaluate performance through cross-modal modeling by assessing text-image relevance or overall semantic consistency (Goyal et al., 2022; Liu et al., 2023; Li et al., 2023). Representative approaches include CLIPScore (Hessel et al., 2021) and LLM-as-a-Judge methods (Anthropic, 2024a,b; Wu et al., 2024; Team, 2025). However, these methods usually generate only overall scores and lack fine-grained interpretability, making it difficult to identify specific errors or support guided refinement (Hessel et al., 2021; Xia et al., 2024).

3 Methodology

In this paper, we propose AgentGER, a agent framework for figure-to-text tasks. Our goal includes two aspects: 1) building a human-aligned and interpretable evaluation method for reliable assessment, and 2) improving summary quality through evaluation-guided feedback, forming an agent framework for constructing high-quality synthetic data. As shown in Figure 2, AgentGER contains three modules. The GenModel uses MLLMs to generate diverse summaries with different quality levels. The EvaModel first learns fine-grained assessment based on human-aligned criteria and introduces a Chain-of-Evaluation (CoE) mechanism to provide interpretable justification for each score. Then, following a curriculum learning strategy, the RefModel further learns evaluation-guided refinement by using CoE feedback to identify weak dimensions and improve the summary accordingly. Through this closed-loop paradigm, AgentGER supports interpretable evaluation and scalable data construction with limited human annotation.

3.1 Generation

The GenModel provides initial figure summaries for subsequent evaluation and refinement. Given an input figure image $\mathbf{I}$, the generation process is defined as:

$$p_{\alpha}(\mathbf{s} \mid \mathbf{I}, q), \quad (1)$$

where $\mathbf{s} = (s_1, s_2, \dots, s_T)$ denotes the generated summary, α denotes the fixed parameters of the generator, and $q \in \{\text{low}, \text{medium}, \text{high}\}$ is the quality-control instruction. GenModel can be built with different MLLMs to increase summary diversity. For each figure, it generates summaries at three quality levels. Low-quality summaries may be vague, incomplete, or inaccurate. Medium-quality summaries cover basic figure information but may miss details or analysis. High-quality summaries aim to faithfully describe the figure, cover key trends and relationships, and provide concise analytical insights. This controllable generation strategy creates diverse candidate summaries and provides the data basis for human annotation, evaluation learning, and refinement training.

3.2 Training Stage 1: Evaluation

The EvaModel aims to provide fine-grained and interpretable quality assessments for figure summaries, aligned with human expert criteria. For each figure image and its corresponding summary, EvaModel evaluates samples across five predefined dimensions: *Faithfulness*, *Completeness*, *Conciseness*, *Logicity*, and *Analysis*. Each dimension employs a discrete scoring range of $\{0, 1, 2\}$, where 0 represents poor, 1 represents acceptable, and 2 represents perfect.

Let $\mathbf{I}$ denotes the input figure image and $\mathbf{s}$ denotes the corresponding summary. Predicting a score vector as: $\mathbf{y} = (y^{(1)}, y^{(2)}, \dots, y^{(5)})$, where $y^{(k)} \in \{0, 1, 2\}$ is the score for the k -th evaluation dimension. The model parameterizes the conditional distribution:

$$p_{\phi}(\mathbf{y} \mid \mathbf{I}, \mathbf{s}), \quad (2)$$

with ϕ denoting the trainable parameters of the model. To enhance interpretability and stability, EvaModel adopts a *Chain-of-Evaluation (CoE)* mechanism. Instead of directly outputting scores, the model is prompted to generate the explicit evaluation reasoning step for each dimension, followed by the corresponding score. Specifically, for each dimension k , the EvaModel provides a reasoning sequence $\mathbf{r}^{(k)}$ and a score $y^{(k)}$ as:

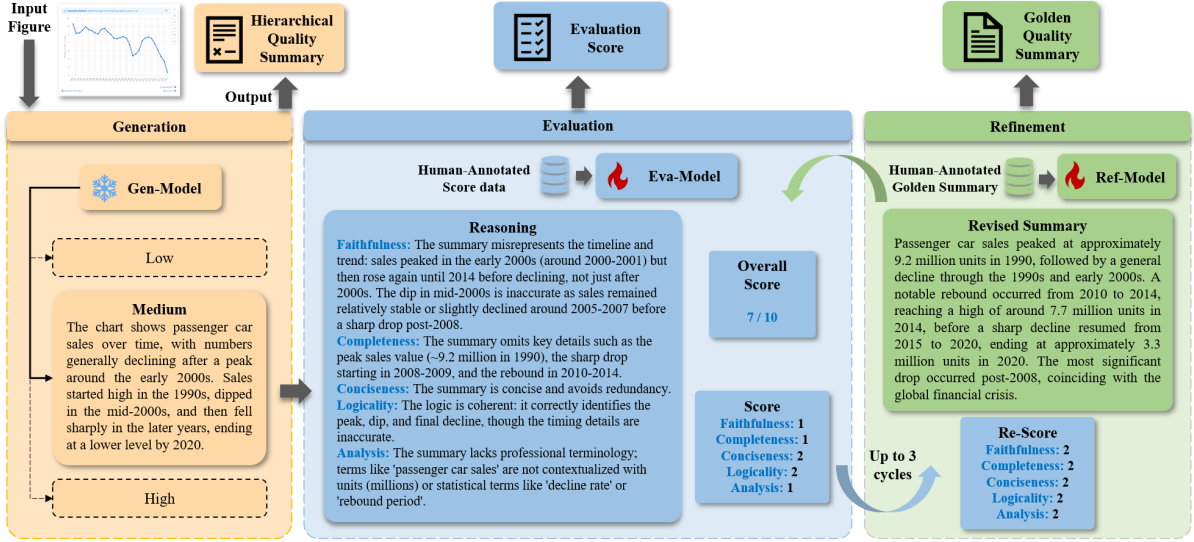


Figure 2: The proposed AgentGER framework.

$$p_{\phi}(\mathbf{r}^{(k)}, y^{(k)} | \mathbf{I}, \mathbf{s}), \quad (3)$$

where $\mathbf{r}^{(k)}$ explains the judgment criteria and evidence used for scoring. This design encourages the model to ground its decisions in observable figure-summary relationships and allows downstream analysis of scoring behavior.

EvaModel is trained on a human-annotated dataset consisting of 1,000 summaries, each labeled with five-dimensional scores with reasoning chains, yielding a total of 5,000 expert annotations. Training is formulated as:

$$\mathcal{L}_{\text{eva}} = -\mathbb{E}_{(\mathbf{I}, \mathbf{s}, \mathbf{z})} \sum_{t=1}^T \log p_{\phi}(z_t | z_{<t}, \mathbf{I}, \mathbf{s}), \quad (4)$$

where $\{z_t\}_{t=1}^T$ represents the token sequence encoding all dimension-wise rationales and scores.

By explicitly modeling multi-dimensional evaluations and their corresponding reasoning chains, EvaModel delivers interpretable assessment signals that align with human assessment criteria. This fine-grained evaluation not only reliably measures summary quality, but also provides structured supervision signals for downstream refinement task.

3.3 Training Stage 2: Evaluation-Guided Refinement

The RefModel extends EvaModel’s capabilities by jointly performing summary evaluation and targeted refinement. RefModel achieves: 1) evaluation performance consistent with EvaModel, and

2) refinement of summaries to address weaknesses identified during evaluation. The training objective is a multi-task, multi-loss optimization. The key challenge lies in enabling refinement without compromising acquired evaluation capabilities.

Let $\mathbf{I}$ denotes the figure image and $\mathbf{s}_{\text{orig}}$ denotes the original summary. RefModel generates a refined summary $\mathbf{s}_{\text{imp}}$ based on the evaluation:

$$p_{\theta}(\mathbf{s}_{\text{imp}} | \mathbf{I}, \mathbf{s}_{\text{orig}}), \quad (5)$$

where θ denotes the trainable parameters of Ref-Model. The refined summary aims to improve low-scoring dimensions while maintaining high-scoring dimensions identified by the evaluation.

To prevent catastrophic forgetting (Luo et al., 2025) of evaluation ability, RefModel is trained using a multi-objective optimization strategy that combines *refinement learning*, *knowledge distillation*, and *experience replay*. The overall training objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{refine}} + \beta \cdot \mathcal{L}_{\text{distill}} + \gamma \cdot \mathcal{L}_{\text{replay}}, \quad (6)$$

where β and γ are weighting coefficients controlling the contribution of each auxiliary loss.

Refinement Loss The refinement loss $\mathcal{L}_{\text{refine}}$ is defined as:

$$\mathcal{L}_{\text{refine}} = -\mathbb{E}_{(\mathbf{I}, \mathbf{s}_{\text{orig}}, \mathbf{s}_{\text{imp}})} \sum_t^T \log p_{\theta}(s_t^{\text{imp}} | s_{<t}^{\text{imp}}, \mathbf{I}, \mathbf{s}_{\text{orig}}), \quad (7)$$

where s_t^{imp} denotes the t -th token of the improved summary.



Figure 3: The examples and statistics of figure types (left) and domains (right) of the proposed FigGER dataset.

Knowledge Distillation Loss To maintain the evaluation capabilities learned from evaluation training stage, RefModel employs a distillation method where the frozen EvaModel serves as the teacher model. Let $\mathbf{z} = (z_1, \dots, z_{|\mathbf{z}|})$ denote the target evaluation output sequence. At each decoding step t , the teacher and student models produce vocabulary-level logits $\mathbf{o}_t^{(\phi)}$ and $\mathbf{o}_t^{(\theta)}$, respectively. The distillation loss is minimized by reducing the Kullback-Leibler (KL) divergence between the output distributions:

$$\mathbf{p}_{T,t}^{(\phi)} = \text{softmax}\left(\frac{\mathbf{o}_t^{(\phi)}}{T}\right), \mathbf{p}_{T,t}^{(\theta)} = \text{softmax}\left(\frac{\mathbf{o}_t^{(\theta)}}{T}\right), \quad (8)$$

$$\mathcal{L}_{\text{distill}} = T^2 \cdot \mathbb{E}_{(\mathbf{I}, \mathbf{s}, \mathbf{z})} \left[\frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \text{KL}(\mathbf{p}_{T,t}^{(\phi)} \parallel \mathbf{p}_{T,t}^{(\theta)}) \right], \quad (9)$$

where $\mathcal{M}$ denotes the set of valid token positions in $\mathbf{z}$, T is the distillation temperature. This term encourages RefModel to preserve the fine-grained scoring knowledge encoded in EvaModel.

Experience Replay Loss To further mitigate forgetting, an experience replay loss $\mathcal{L}_{\text{replay}}$ is applied on the evaluation data:

$$\mathcal{L}_{\text{replay}} = -\mathbb{E}_{(\mathbf{I}, \mathbf{s}_{\text{orig}}, \mathbf{z})} \sum_t^T \log p_{\theta}(z_t \mid z_{<t}, \mathbf{I}, \mathbf{s}_{\text{orig}}), \quad (10)$$

where z denotes the token sequence in the samples of evaluation task.

By jointly optimizing these objectives, RefModel learns to generate higher-quality summaries

while maintaining evaluation consistency with EvaModel. This design enables AgentGER to iteratively refine summaries in a stable and controllable way, forming an effective closed-loop refinement process.

4 Dataset

We construct a hybrid annotation dataset combining human experts and MLLMs for the figure-to-text task called FigGER. Source figures are sampled from publicly web platforms and existing open-source datasets, covering 10 figure types and 27 domains (as shown in Figure 3) to ensure diversity in structural formats and application domains. As shown in Table 1, the dataset comprises 11,000 figure summary texts and 55,000 dimensional scoring labels with reasoning chains. Specifically: 1) 1,000 high-quality seed data samples annotated by human experts are used for model training; 2) 10,000 model-generated synthetic data samples are employed to expand the dataset scale; 3) 1,000 samples selected through stratified sampling from 10,000 synthetic data undergo manual verification for quality checks and benchmark construction. This dataset is constructed through a multi-stage process encompassing model generation, human annotation, synthetic-scaled data, and human verification, balancing both data quality and scalability.

Figure Collection Source figures are collected from publicly available web platforms and existing figure understanding datasets, spanning diverse application domains including scientific research, finance, and statistical analysis, etc. Figure types are initially filtered to retain only structured figures (such as bar charts, line charts, scatter plots, etc.), thereby avoiding interference from illustra-

	Subset	Ann.	Sample	Label
Human	Seed Training	Human	1000	5000
Synthetic	Model-Generated	Model	9000	45000
	Golden Test	Human	1000	5000
Total			11000	55000

Table 1: Statistics of the FigGER. Ann.=Annotation.

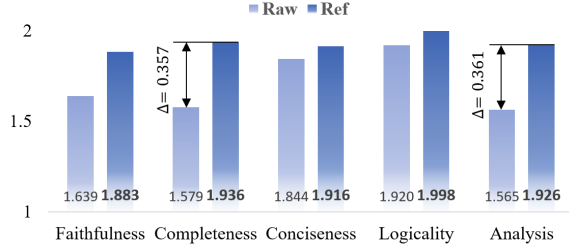


Figure 4: Comparison of average scores across all dimensions for test set samples with (Ref) and without (Raw) refinement.

tions, flowcharts, or unstructured visual content that could confuse task definitions. Then, a combined manual and rule-based filtering process is applied to assess figure resolution, legibility, and informational completeness. Samples exhibiting missing axis labels, blurred images, or incomplete data representation are removed to ensure the feasibility and reliability of subsequent summarization and evaluation tasks. Details in Appendix A.1.

Data Annotation Annotation is completed through a collaborative approach combining model generation, manual annotation, and human refinement. First, MLLMs automatically generate summary texts of different quality for figures. Then, 8 annotators with relevant expertise manually evaluate these summaries. We assign discrete score labels $[0, 1, 2]$ across five dimensions: faithfulness, completeness, conciseness, logicity, and analysis. Each summary corresponds to five dimensional scores with reasoning chains, forming fine-grained, multi-dimensional quality annotations. Upon this foundation, annotators further manually refine the summaries to generate high-quality golden summaries. The final 1,000 manually annotated and refined high-quality data is used to train the EvaModel and RefModel, ensuring model assessment aligns with human expert criteria. More details in Appendix A.2 and A.3.

Synthetic Data Quality Assessment To evaluate the quality of synthetic data, we conduct human verification on 1,000 samples selected through

stratified sampling from 10,000 synthetic samples. Each sample is independently annotated by two annotators, and disagreements are resolved by a senior annotator. The human annotations show high consistency, with $Pearson=0.758$ and $Spearman=0.801$. We further compare the scores generated by AgentGER with human annotations on this verified subset. The results show (Table 2) strong agreement between AgentGER and human experts ($Pearson=0.747$ and $Spearman=0.776$), indicating that the synthetic scores can provide reliable supervision signals for training and evaluation. These 1,000 human-verified samples are also used as the benchmark test set for comparing different evaluation methods. In addition to evaluation reliability, we verify the quality of refinement. As shown in Figure 4, summaries refined by RefModel obtain higher scores than the initial generated summaries across all dimensions. Detailed results in Table 4 further confirm that the refined summaries consistently improve under AgentGER-based, independent model-based, and human evaluation. These results suggest that our synthetic data not only provides human-aligned evaluation, but also leads to real quality improvements to achieve expert-level figure summaries. Details in Appendix A.4.

5 Experimental setup

Baselines: We conduct experiments using 4 reference-based and 9 reference-free methods. We also introduce the consistency scores from independent human annotators as the upper bound. **Reference-based:** BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), BERTScore (Zhang et al., 2019), CIDEr (Vedantam et al., 2015). **Reference-free:** UniChart (Masry et al., 2023), ChartInstruct (Masry et al., 2024), Qwen2.5-VL-7B (Team, 2025), Qwen3-VL-8B, Gemini-2.5-Flash (Comanici et al., 2025), Gemini-3-Pro (Comanici et al., 2025), GPT-5-Mini (Singh et al., 2025), MiniCPM-V2.6 (Yao et al., 2024), InterVL-2.5-8B (Chen et al., 2024). Details in Appendix B.1.

Dataset and Evaluation: We evaluate all methods on the FigGER golden test set using four metrics: Pearson Correlation (PC), Spearman Correlation (SC), Mean Absolute Error (MAE), and Mean Squared Error (MSE). Details in Appendix B.2.

Settings: We use the Qwen3-VL-8B-Instruct model as the backbone model and adopt Lora (Hu et al., 2022) for training, with rank $r = 64$,

scaling factor $\alpha = 128$, and dropout rate 0.05. AdamW (Loshchilov and Hutter, 2017) is the optimizer with a learning rate of 2×10^{-4} . $\beta = 0.5$, $\gamma = 0.3$. Detailed instructions and scoring criteria are shown in the Appendix B.3. For proprietary models, we use official APIs. For open-source models, we use the official GitHub or Huggingface code and reproduce them using default parameters. Experiments on 8 NVIDIA-H800 GPUs, and we report average results over 5 runs.

6 Results

6.1 Main Results

Table 2 reports the overall evaluation results:

Reference-based methods show poor performance. Whether using n-gram matching methods (such as BLEU and ROUGE) or BERTScore incorporating semantic embeddings, their consistency with human evaluation remains low. This indicates that solely relying on shallow semantic similarity between model-generated and reference summaries fails to accurately reflect the true quality of figure summaries under human-aligned criteria such as faithfulness, logicity, and analysis.

Reference-free MLLMs perform better, but black-box scoring still falls short of human alignment. Compared with reference-based metrics, MLLM-based evaluators achieve higher correlations with human judgments. Stronger models such as Qwen3-VL-8B and Gemini-3-Pro obtain better PC and SC scores, showing the benefit of stronger cross-modal understanding. However, Gemini-3-Pro still shows a clear gap compared with human performance, with a PC of 0.639 and an SC of 0.629, compared with the human PC of 0.758 and SC of 0.801. This suggests that existing reference-free methods remain limited when used as black-box evaluators without task-specific alignment.

Training improves evaluation performance, while multi-dimensional CoE further strengthens human alignment. The trained Qwen3-VL-8B baseline, which directly predicts an overall score without multi-dimensional evaluation or CoE, improves over its reference-free counterpart, reaching 0.643 PC and 0.639 SC. AgentGER w/o CoE further improves the results, but still remains clearly below the full AgentGER. This shows that task-specific training is useful, while the Chain-of-Evaluation mechanism provides additional gains by making the evaluation process more structured and aligned with human criteria.

	PC(↑)	SC(↑)	MAE(↓)	MSE(↓)
Reference-based Methods				
Human	0.758	0.801	0.096	0.065
BLEU	0.244	0.277	0.123	0.023
ROUGE	0.367	0.323	0.244	0.084
BERTScore	0.363	0.225	0.728	0.531
CIDEr	0.105	-0.011	0.367	0.554
Reference-free Methods				
InternVL2.5-8B	0.391	0.364	0.227	0.129
MiniCPM-V2.6	0.475	0.345	0.191	0.092
GPT-5-Mini	0.469	0.425	0.385	0.237
Gemini-2.5-Flash	0.505	0.462	0.396	0.302
Qwen2.5-VL-7B	0.533	0.542	0.213	0.132
Qwen3-VL-8B	0.595	0.609	0.126	0.081
Gemini-3-Pro	0.639	0.629	0.394	0.269
Trained Methods				
ChartInstruct	0.421	0.430	0.187	0.113
UniChart	0.448	0.433	0.185	0.104
Qwen3-VL-8B	0.643	0.639	0.135	0.109
AgentGER w/o COE	0.651	0.642	0.171	0.124
AgentGER w/o KD	0.686	0.668	0.143	0.097
AgentGER w/o ER	0.697	0.644	0.124	0.066
AgentGER(8B)	0.747	0.776	0.085	0.057

Table 2: Main evaluation results. KD=Knowledge Distillation Loss. ER=Experience Replay Loss.

AgentGER achieves the best performance across all metrics and approaches human-level agreement. AgentGER obtains the highest PC of 0.747 and SC of 0.776, while also achieving the lowest MAE and MSE among all methods. It outperforms both general MLLMs and trained models, showing the effectiveness of human-aligned multi-dimensional evaluation. The ablation results further show that removing knowledge distillation (KD) or experience replay (ER) degrades performance, confirming that KD and ER help preserve evaluation consistency and stabilize refinement-oriented training.

6.2 Breakdown Analysis

Table 3 presents results across the five evaluation dimensions. Existing baselines show clear imbalance in fine-grained evaluation performance. Most models achieve relatively strong results on *Completeness*, but perform much worse on *Conciseness* and *Logicity*. The strongest Gemini-3-Pro obtains negative correlations on *Conciseness* (PC=-0.096 and SC=-0.081). Although larger models generally improve overall performance, the gains remain limited and inconsistent across dimensions, indicating that increasing model scale alone cannot effectively improve human-aligned fine-grained evaluation.

Model	Faithfulness		Completeness		Conciseness		Logicity		Analysis		Overall	
	PC(↑)	SC(↑)	PC(↑)	SC(↑)	PC(↑)	SC(↑)	PC(↑)	SC(↑)	PC(↑)	SC(↑)	PC(↑)	SC(↑)
InternVL2.5-8B	0.183	0.191	0.416	0.385	0.008	0.012	0.228	0.256	0.294	0.289	0.391	0.364
MiniCPM-V2.6	0.331	0.337	0.343	0.248	0.101	0.097	0.581	0.687	0.341	0.231	0.475	0.345
GPT-5-Mini	0.187	0.183	0.489	0.467	0.269	0.332	0.051	0.079	0.432	0.416	0.469	0.425
Gemini-2.5-Flash	0.284	0.271	0.503	0.495	0.026	0.061	0.197	0.198	0.447	0.429	0.505	0.462
Qwen2.5-VL-7B	0.381	0.391	0.516	0.481	0.308	0.311	0.408	0.435	0.482	0.439	0.533	0.542
Qwen3-VL-8B	0.531	0.542	0.599	0.592	0.599	0.653	0.593	0.766	0.584	0.586	0.595	0.609
Gemini-3-Pro	0.421	0.392	0.666	0.677	-0.096	-0.081	0.188	0.223	0.680	0.687	0.639	0.629
AgentGER (8B)	0.711	0.742	0.704	0.718	0.713	0.744	0.699	0.878	0.712	0.731	0.747	0.776

Table 3: Breakdown analysis results of five-dimensions on the evaluation task.

	Faith.	Compl.	Conci.	Logic.	Analy.
<i>AgentGER Evaluation on Full Test Set</i>					
Gemini-3-Pro*	1.711	1.689	1.856	1.915	1.689
Human*	1.958	1.931	1.936	1.965	1.956
w/o Ref	1.639	1.579	1.844	1.920	1.565
Δ	↑0.244	↑0.357	↑0.072	↑0.078	↑0.361
AgentGER*	1.883	1.936	1.916	1.998	1.926
<i>Gemini-2.5-Flash Evaluation on Full Test Set</i>					
Gemini-3-Pro*	1.726	1.668	1.861	1.923	1.703
Human*	1.878	1.865	1.856	1.917	1.901
w/o Ref	1.602	1.548	1.806	1.874	1.512
Δ	↑0.210	↑0.303	↑0.073	↑0.063	↑0.330
AgentGER*	1.812	1.851	1.879	1.937	1.842
<i>Human Evaluation on 200 Sampled Examples</i>					
Gemini-3-Pro*	1.724	1.671	1.868	1.936	1.705
Human*	1.915	1.881	1.903	1.951	1.932
w/o Ref	1.655	1.590	1.852	1.912	1.580
Δ	↑0.215	↑0.315	↑0.058	↑0.043	↑0.322
AgentGER*	1.870	1.905	1.910	1.955	1.902

Table 4: Refinement performance. “w/o Ref”= AgentGER without refinement. Marked * = after refinement.

AgentGER achieves the best and balanced performance across all dimensions. It obtains strong results on foundational dimensions such as *Faithfulness* (PC=0.711) and *Completeness* (PC=0.704), while also maintaining clear advantages on more challenging dimensions including *Conciseness* (PC=0.713) and *Logicity* (PC=0.699). These results demonstrate that the combination of multi-dimensional human-aligned evaluation and the Chain-of-Evaluation mechanism enables more stable and interpretable assessment than existing overall-scoring methods, providing reliable guidance for downstream summary refinement.

6.3 Refinement Performance

Based on the strong evaluation capability of AgentGER, we further examine whether its evaluation feedback can effectively guide summary refinement. To avoid overestimating refinement quality using only the internal evaluator, we report three

types of results in Table 4: 1) AgentGER-based evaluation on the full test set, 2) independent evaluation using Gemini-2.5-Flash on the full test set, and 3) human evaluation on a randomly sampled subset of 200 examples.

As shown in Table 4, refined summaries consistently outperform the original summaries across all five dimensions under all evaluation settings. The largest improvements are observed on *Completeness* and *Analysis*, indicating that AgentGER effectively enhances missing information and analytical depth. The improvements remain stable under both independent and human evaluation, suggesting that the gains are not caused by internal evaluator preference. We also select the strongest Gemini-3-Pro and human to refine summaries, AgentGER achieves consistently higher scores across all dimensions and evaluation settings compared with Gemini-3-Pro, demonstrating that AgentGER generates effective feedback for targeted summary refinement and achieves human-level performance.

7 Conclusion

We introduce AgentGER, a unified agent framework for figure-to-text that jointly supports quality-aware generation, human-aligned interpretable evaluation, and target-guided refinement. AgentGER enables scalable construction of high-quality figure summaries and reliable evaluation aligned with human expert criteria. Extensive experiments demonstrate that AgentGER consistently outperforms strong baseline models on evaluation benchmarks and achieves human-level performance on both evaluation and refinement tasks. These results highlight the effectiveness of integrating multi-dimensional Chain-of-Evaluation and target-guided refinement, and suggest that this new paradigm offer a promising direction for human-aligned and robust figure understanding systems.

Limitations

Although AgentGER shows strong performance on figure-to-text evaluation and refinement, the current task mainly focuses on figure understanding. In broader domains, such as scientific, financial, and medical report summarization, evaluation and refinement also require substantial expert annotations. To the best of our knowledge, there are still few publicly available expert-level datasets for these tasks. Therefore, extending AgentGER to these high-value domains and constructing reliable expert-aligned benchmarks will be an important direction for future work.

References

- Mubashara Akhtar, Nikesh Subedi, Vivek Gupta, Sahar Tahmasebi, Oana Cocarascu, and Elena Simperl. 2024. Chartcheck: Explainable fact-checking over real-world chart images. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 13921–13937.
- Anthropic. 2024a. The claude 3 model family: Opus, sonnet, haiku.
- Anthropic. 2024b. Model card addendum: Claude 3.5 haiku and upgraded claude 3.5 sonnet.
- Anthropic. 2025. Claude 3.7 sonnet and claude code. Accessed: 2025-05-20.
- Ritwick Chaudhry, Sumit Shekhar, Utkarsh Gupta, Pranav Maneriker, Prann Bansal, and Ajay Joshi. 2020. Leaf-qa: Locate, encode & attend for figure question answering. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3512–3521.
- Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, and 1 others. 2024. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24185–24198.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, and 1 others. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*.
- Kenny Davila, Srirangaraj Setlur, David Doermann, Bhargava Urala Kota, and Venu Govindaraju. 2021. Chart mining: A survey of methods for automated chart analysis. *IEEE transactions on pattern analysis and machine intelligence*, 43(11):3799–3819.
- Alexander R Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. 2021. Summeval: Re-evaluating summarization evaluation. *Transactions of the Association for Computational Linguistics*, 9:391–409.
- Théo Gigant, Camille Guinaudeau, Marc Decombas, and Frédéric Dufaux. 2024. Mitigating the impact of reference quality on evaluation of summarization systems with reference-free metrics. *arXiv preprint arXiv:2410.10867*.
- Tanya Goyal, Junyi Jessy Li, and Greg Durrett. 2022. News summarization and evaluation in the era of gpt-3. *arXiv preprint arXiv:2209.12356*.
- Yucheng Han, Chi Zhang, Xin Chen, Xu Yang, Zhibin Wang, Gang Yu, Bin Fu, and Hanwang Zhang. 2023a. Chartllama: A multimodal llm for chart understanding and generation. *arXiv preprint arXiv:2311.16483*.
- Yucheng Han, Chi Zhang, Xin Chen, Xu Yang, Zhibin Wang, Gang Yu, Bin Fu, and Hanwang Zhang. 2023b. Chartllama: A multimodal llm for chart understanding and generation. *Preprint*, arXiv:2311.16483.
- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. 2021. Clipscore: A reference-free evaluation metric for image captioning. *arXiv preprint arXiv:2104.08718*.
- Ting-Yao Hsu, C Lee Giles, and Ting-Hao’Kenneth’ Huang. 2021. Scicap: Generating captions for scientific figures. *arXiv preprint arXiv:2110.11624*.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, and 1 others. 2022. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3.
- Chandra Irugalbandara, Ashish Mahendra, Roland Daynauth, Tharuka Kasthuri Arachchige, Jayanaka Dantanarayana, Krisztian Flautner, Lingjia Tang, Yiping Kang, and Jason Mars. 2024. Scaling down to scale up: A cost-benefit analysis of replacing openai’s llm with open source slms in production. In *2024 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, pages 280–291. IEEE.
- Kushal Kafle, Brian Price, Scott Cohen, and Christopher Kanan. 2018. Dvqa: Understanding data visualizations via question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5648–5656.
- Samira Ebrahimi Kahou, Vincent Michalski, Adam Atkinson, Ákos Kádár, Adam Trischler, and Yoshua Bengio. 2017. Figureqa: An annotated figure dataset for visual reasoning. *arXiv preprint arXiv:1710.07300*.
- Shankar Kantharaj, Rixie Tiffany Ko Leong, Xiang Lin, Ahmed Masry, Megh Thakkar, Enamul Hoque, and Shafiq Joty. 2022. Chart-to-text: A large-scale

- benchmark for chart summarization. *arXiv preprint arXiv:2203.06486*.
- Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David Shamma, and 1 others. 2017. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *International Journal of Computer Vision*, 123(1).
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR.
- Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft coco: Common objects in context. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13*, pages 740–755. Springer.
- Fuxiao Liu, Xiaoyang Wang, Wenlin Yao, Jianshu Chen, Kaiqiang Song, Sangwoo Cho, Yaser Yacoob, and Dong Yu. 2024. Mmc: Advancing multimodal chart understanding with large-scale instruction tuning. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1287–1310.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-eval: Nlg evaluation using gpt-4 with better human alignment. *arXiv preprint arXiv:2303.16634*.
- Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*.
- Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. 2025. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *IEEE Transactions on Audio, Speech and Language Processing*.
- Ahmed Masry, Parsa Kavehzadeh, Xuan Long Do, Enamul Hoque, and Shafiq Joty. 2023. Unichart: A universal vision-language pretrained model for chart comprehension and reasoning. *arXiv preprint arXiv:2305.14761*.
- Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. *arXiv preprint arXiv:2203.10244*.
- Ahmed Masry, Mehrad Shahmohammadi, Md Rizwan Parvez, Enamul Hoque, and Shafiq Joty. 2024. Chartinstruct: Instruction tuning for chart comprehension and reasoning. *arXiv preprint arXiv:2403.09028*.
- Fanqing Meng, Wenqi Shao, Quanfeng Lu, Peng Gao, Kaipeng Zhang, Yu Qiao, and Ping Luo. 2024. Chartassistant: A universal chart multimodal language model via chart-to-table pre-training and multitask instruction tuning. *arXiv preprint arXiv:2401.02384*.
- Nitesh Methani, Pritha Ganguly, Mitesh M Khapra, and Pratyush Kumar. 2020. Plotqa: Reasoning over scientific plots. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 1527–1536.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318.
- Raian Rahman, Rizvi Hasan, Abdullah Al Farhad, Md Tahmid Rahman Laskar, Md Hamjajul Ashmafee, and Abu Raihan Mostofa Kamal. 2023. Chartsumm: A comprehensive benchmark for automatic chart summarization of long and short summaries. *arXiv preprint arXiv:2304.13620*.
- Aaditya Singh, Adam Fry, Adam Perelman, Adam Tart, Adi Ganesh, Ahmed El-Kishky, Aidan McLaughlin, Aiden Low, AJ Ostrow, Akhila Ananthram, and 1 others. 2025. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267*.
- Hrituraj Singh and Sumit Shekhar. 2020. Stl-cqa: Structure-based transformers with localization and encoding for chart question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3275–3284.
- Qwen Team. 2025. [Qwen2.5 technical report](#). *Preprint*, arXiv:2412.15115.
- Ramakrishna Vedantam, C Lawrence Zitnick, and Devi Parikh. 2015. Cider: Consensus-based image description evaluation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4566–4575.
- Zirui Wang, Mengzhou Xia, Luxi He, Howard Chen, Yitao Liu, Richard Zhu, Kaiqu Liang, Xindi Wu, Haotian Liu, Sadhika Malladi, and 1 others. 2025. Charxiv: Charting gaps in realistic chart understanding in multimodal llms. *Advances in Neural Information Processing Systems*, 37:113569–113697.
- Zhiyu Wu, Xiaokang Chen, Zizheng Pan, Xingchao Liu, Wen Liu, Damai Dai, Huazuo Gao, Yiyang Ma, Chengyue Wu, Bingxuan Wang, and 1 others. 2024. Deepseek-vl2: Mixture-of-experts vision-language models for advanced multimodal understanding. *arXiv preprint arXiv:2412.10302*.

- Renqiu Xia, Bo Zhang, Haoyang Peng, Hancheng Ye, Xiangchao Yan, Peng Ye, Botian Shi, Yu Qiao, and Junchi Yan. 2023. Structchart: Perception, structuring, reasoning for visual chart understanding. *arXiv preprint arXiv:2309.11268*.
- Renqiu Xia, Bo Zhang, Hancheng Ye, Xiangchao Yan, Qi Liu, Hongbin Zhou, Zijun Chen, Min Dou, Botian Shi, Junchi Yan, and 1 others. 2024. Chartx & chartvlm: A versatile benchmark and foundation model for complicated chart reasoning. *arXiv preprint arXiv:2402.12185*.
- Zhengzhuo Xu, Sinan Du, Yiyan Qi, Chengjin Xu, Chun Yuan, and Jian Guo. 2023. Chartbench: A benchmark for complex visual reasoning in charts. *arXiv preprint arXiv:2312.15915*.
- Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, and 1 others. 2024. Minicpm-v: A gpt-4v level mllm on your phone. *arXiv preprint arXiv:2408.01800*.
- Tan Yue, Xuzhao Shi, Rui Mao, Zilong Song, Zonghai Hu, and Dongyan Zhao. 2025. Anafig: A human-aligned dataset for scientific figure analysis. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 12837–12843.
- Duzhen Zhang, Yahan Yu, Jiahua Dong, Chenxing Li, Dan Su, Chenhui Chu, and Dong Yu. 2024a. Mmllms: Recent advances in multimodal large language models. *arXiv preprint arXiv:2401.13601*.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. 2019. Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*.
- Yue Zhang, Ming Zhang, Haipeng Yuan, Shichun Liu, Yongyao Shi, Tao Gui, Qi Zhang, and Xuanjing Huang. 2024b. Llmeval: A preliminary study on how to evaluate large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19615–19622.

A Dataset Details

We provide additional details about the construction of FigGER. These details are intended to improve the reproducibility of the dataset construction process and further clarify how data quality is controlled during human-machine collaborative annotation.

A.1 Figure Collection and Filtering

The source figures in FigGER are collected from publicly available web data and existing open-source figure understanding datasets. To ensure that the collected figures are suitable for the figure-to-text task, we only retain structured data visualizations that contain explicit visual encodings, such as axes, legends, data points, bars, lines, sectors, or other chart elements. Figures that mainly serve as illustrations, flowcharts, conceptual diagrams, natural images, or screenshots are excluded, since they do not match the task definition of scientific figure summarization.

After the initial collection, we conduct a two-stage filtering process. First, rule-based filtering is used to remove figures with extremely low resolution, abnormal aspect ratios, unreadable text, or corrupted image files. Second, manual inspection is performed to further check whether each figure contains sufficient information for summary generation and quality evaluation. During this process, figures are removed if they have missing or unreadable axis labels, incomplete legends, severe blur, excessive occlusion, or insufficient visual evidence to support a meaningful summary. We also remove duplicate or near-duplicate figures to reduce data redundancy and avoid potential leakage across different subsets.

The final figure pool covers multiple figure types and application domains. This design ensures that FigGER contains diverse visual structures and domain contexts, allowing the evaluation and refinement models to learn generalizable figure understanding and summarization abilities rather than overfitting to a narrow chart format.

A.2 Annotation Guidelines

The annotation guidelines are designed to support fine-grained and interpretable evaluation of figure summaries. Each summary is evaluated along five dimensions: *Faithfulness*, *Completeness*, *Conciseness*, *Logicality*, and *Analysis*. Each dimension is assigned a discrete score from $\{0, 1, 2\}$, where 0 in-

icates poor quality, 1 indicates acceptable quality, and 2 indicates high quality.

Figure 5 presents the detailed scoring criteria used by annotators. In general, *Faithfulness* measures whether the summary is consistent with the visual evidence in the figure and whether it avoids unsupported claims. *Completeness* measures whether the summary covers the main visual elements, trends, comparisons, and key findings. *Conciseness* measures whether the summary expresses the necessary information without redundancy or irrelevant content. *Logicality* measures whether the summary is coherent, well organized, and free from contradictory statements. *Analysis* measures whether the summary goes beyond surface-level description and provides meaningful interpretation of trends, relationships, or implications.

For each dimension, annotators are required to provide both a score and a short reasoning explanation. The reasoning explanation should identify the main evidence supporting the score, such as correctly described trends, missing information, factual errors, redundant expressions, or weak analytical depth. This design generates both numerical labels and Chain-of-Evaluation rationales, which are used to train the evaluation model to make interpretable and human-aligned judgments.

A.3 Human Annotation Protocol

The human annotation process is conducted by annotators with experience in scientific figure reading, data visualization, or natural language annotation. All annotators hold undergraduate and master’s degrees from top-tier universities, and while being native Chinese speakers, they possess excellent proficiency in English. Before formal annotation, all annotators are trained with the annotation guidelines and scoring criteria. They first annotate a small set of trial examples, after which disagreements are discussed to unify the interpretation of each scoring dimension. This training stage helps reduce subjective bias and improves consistency across annotators. The average hourly payment for each annotator is 150 CNY, exceeding the minimum wage. Additionally, annotators are given a 10-minute break after 30 minutes of work. The dataset and code will be open-sourced under the MIT License.

During formal annotation, each sample is evaluated according to the five-dimensional criteria. Annotators first inspect the figure and the corresponding summary, then assign scores for all five

Faithfulness	Check whether the summary strictly adheres to the figure's facts, including values, trends, proportions, and labels; without fabrication, distortion, or misinterpretation. ➤ Score 0: The summary severely deviates from the figure's facts, with errors or fabricated content. ➤ Score 1: Largely consistent with the figure, but with partial deviations or inaccurate details. ➤ Score 2: Fully faithful to the figure's facts, with all information correct.
Completeness	Check whether the summary covers core elements, main trends, and key comparisons in the figure. ➤ Score 0: Misses major content or key information. ➤ Score 1: Covers the main information but with secondary omissions. ➤ Score 2: Fully covers all key information.
Conciseness	Check whether the summary is compact, information-dense, and free of redundancy. ➤ Score 0: Redundant, or overly brief to the point of missing key information. ➤ Score 1: Generally concise but can be further optimized. ➤ Score 2: Tersely written with efficient expression.
Logicity	Check whether the logic is coherent, sequence is reasonable, and causal relations are clear. ➤ Score 0: Logical confusion or self-contradiction. ➤ Score 1: Basically coherent but with slight jumps or ambiguity. ➤ Score 2: Clear logic and strong consistency.
Analysis	Check whether the summary meets domain-standard expression (e.g., finance, statistics, economics). ➤ Score 0: Lacks professionalism, or uses incorrect terminology/units. ➤ Score 1: Professional expression is insufficient or slightly colloquial. ➤ Score 2: Accurate terminology, formal tone, and analytical depth.

Figure 5: Five-dimensional scoring criteria used in human annotation.

dimensions and write dimension-specific reasoning explanations. For the human-verified golden test set, each sample is independently annotated by two annotators. If the two annotators provide inconsistent scores or conflicting reasoning, the sample is reviewed by a senior annotator, who makes the final decision after checking the figure, summary, and both annotation records.

In addition to scoring, annotators also refine the original summaries to construct high-quality golden summaries. During refinement, annotators are asked to preserve correct information, remove unsupported or redundant content, correct factual errors, and add missing key trends or analytical observations when necessary. The refined summaries are further checked to ensure that they are faithful to the figure, complete in content, concise in expression, logically organized, and analytically meaningful. Figure 6 shows the annotation interface used in our annotation process.

A.4 Synthetic Data Generation and Verification

To expand the dataset while maintaining annotation quality, we adopt a human-machine collaborative strategy for synthetic data generation. Starting

from the human-annotated seed data, the evaluation model and refinement model are trained to perform fine-grained scoring and targeted summary improvement. The trained models are then used to generate large-scale synthetic samples, including figure summaries, five-dimensional scores, and Chain-of-Evaluation rationales.

The synthetic summaries are generated with different quality levels to simulate the diverse quality distribution observed in real model outputs. Specifically, low-quality summaries may contain vague descriptions, missing key information, factual errors, or weak logical structure. Medium-quality summaries usually cover the main content of the figure but may still miss secondary information, contain minor redundancy, or lack analytical depth. High-quality summaries are expected to accurately describe the figure, cover important trends and comparisons, and provide concise analytical insights. Figure 7 shows examples of summaries at different quality levels.

To verify the quality of synthetic data, we construct a human-verified subset from the synthetic pool. Instead of relying only on uncontrolled random inspection, we ensure that the verified sub-

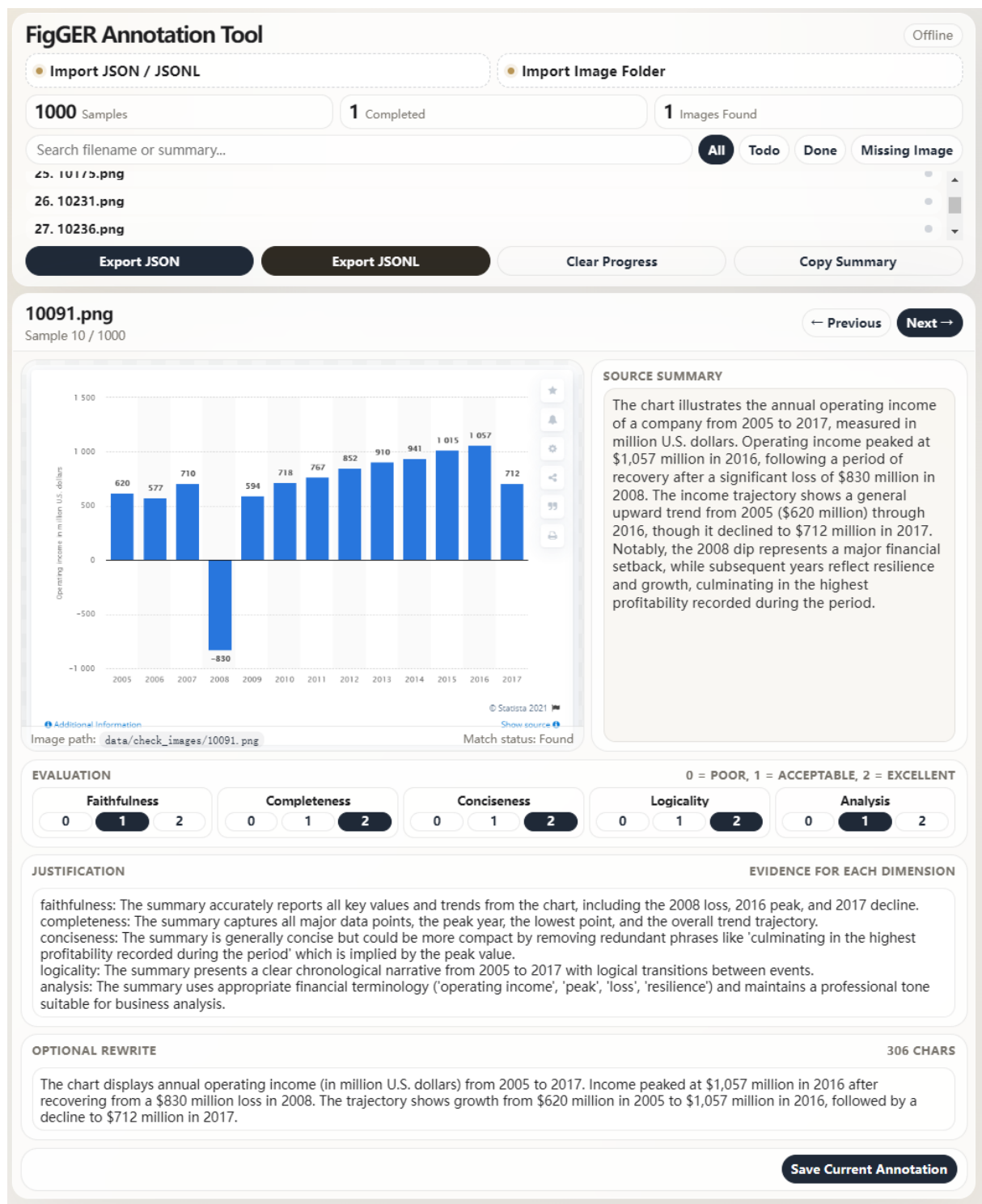


Figure 6: Annotation tool used for five-dimensional scoring and reasoning annotation.

set covers diverse figure types, domains, predicted score ranges, and quality levels. This verification strategy allows us to examine whether the synthetic data remain reliable across different visual structures and summary qualities. Each selected sample is re-annotated by human annotators using the same five-dimensional criteria. The human annotations

are then compared with model-generated scores to measure the consistency between synthetic labels and expert judgments.

The verification results show that the synthetic labels are highly consistent with human evaluation. In the main experiments, AgentGER achieves strong agreement with human annotations, with an

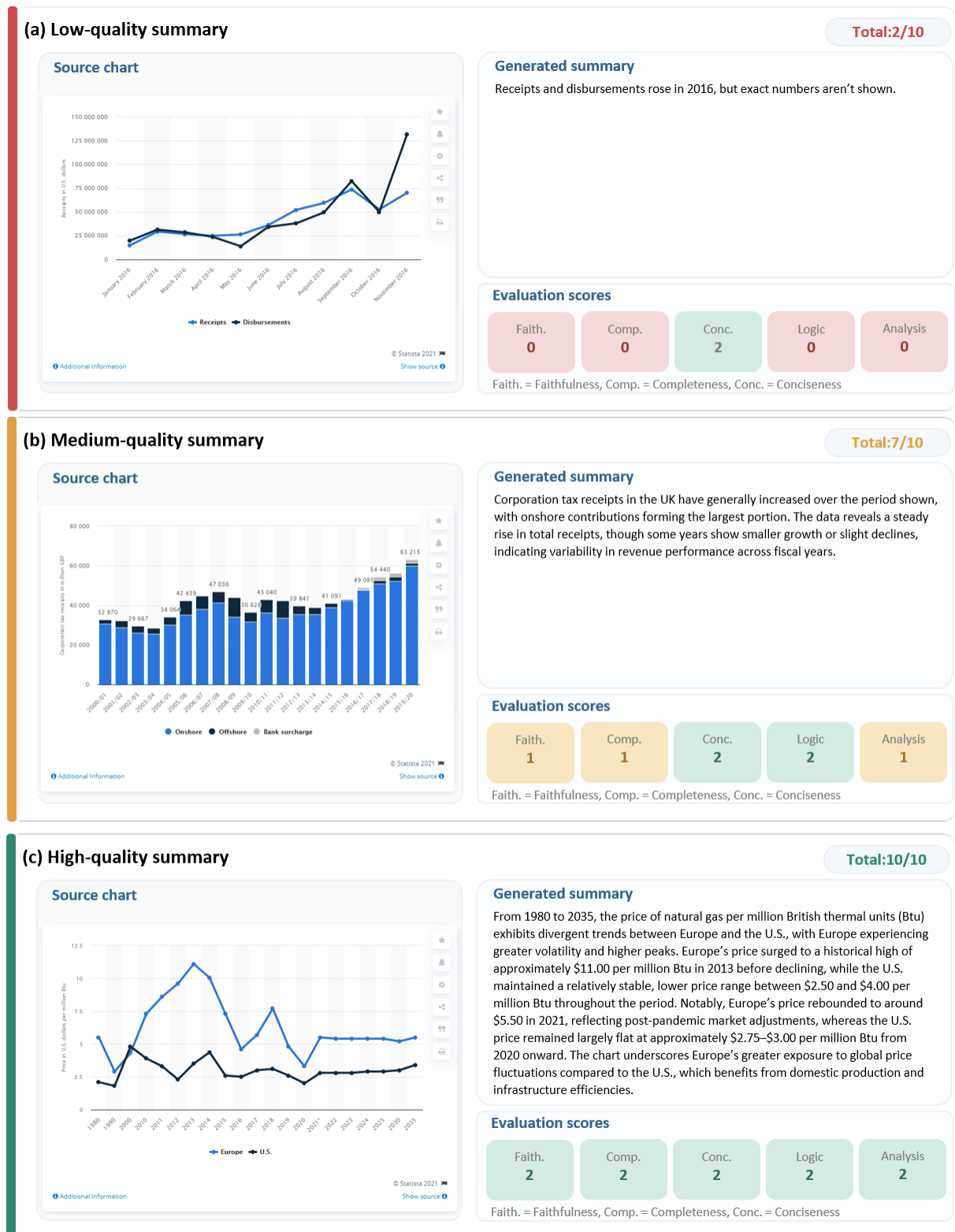


Figure 7: Examples of generated summaries with different quality levels.

overall Pearson correlation of 0.747 and Spearman correlation of 0.776. These results are close to the human agreement level, indicating that the generated scores can serve as reliable supervision signals for training and evaluation. Moreover, the refine-

ment results further show that summaries improved by the refinement model receive higher scores under both model-based and human evaluation, suggesting that the synthetic refinement process generates real quality improvements rather than only

optimizing toward the internal evaluator.

A.5 Data Splits and Leakage Control

FigGER contains three subsets: the human seed training set, the synthetic training set, and the human-verified golden test set. The human seed training set contains 1,000 samples manually annotated by experts. These samples are used to train the initial evaluation model and refinement model. The synthetic training set contains 9,000 model-generated samples, which are used to expand the training data and improve the scalability of the framework. The human-verified golden test set contains 1,000 samples selected from the synthetic pool and re-annotated by human experts.

To avoid data leakage, we apply split-level isolation throughout dataset construction and model training. First, samples in the human-verified golden test set are excluded from all training stages. Second, duplicate and near-duplicate figures are removed before constructing the final splits, reducing the possibility that visually identical or highly similar figures appear in both training and test sets. Third, the golden test labels are produced through independent human verification and are not exposed to the model during training. These controls ensure that the reported evaluation results reflect generalization to unseen test samples rather than memorization of training data. Table 5 summarizes the data split used in FigGER.

B Experimental Setup Details

This appendix provides additional details of the experimental setup, including the baseline methods and evaluation metrics used in the main experiments.

B.1 Baselines

We compare AgentGER with both reference-based and reference-free evaluation methods. Reference-based methods evaluate a generated summary by comparing it with a human-written reference summary. Reference-free methods directly assess the quality of a summary based on the input figure and the generated text, without requiring a reference summary. We also report the consistency between independent human annotators as a human-level reference.

Human. We use independent human annotations as an upper-bound reference for evaluation consistency. Specifically, each sample in the human-verified golden test set is independently annotated

by human experts, and the agreement between annotators is measured using the same metrics as model evaluation.

BLEU (Papineni et al., 2002) is a widely used automatic evaluation metric originally proposed for machine translation. It measures the n-gram precision between a candidate text and one or more reference texts, with a brevity penalty to discourage overly short outputs.

ROUGE (Lin, 2004) is a standard evaluation package for text summarization. It measures the lexical overlap between generated and reference summaries, including variants based on n-gram overlap and longest common subsequence matching.

BERTScore (Zhang et al., 2019) evaluates text generation by computing token-level semantic similarity between candidate and reference texts using contextual embeddings from pretrained language models. Compared with exact lexical matching, it is more tolerant to paraphrases and semantic variations.

CIDEr (Vedantam et al., 2015) was originally proposed for image description evaluation. It measures the similarity between generated captions and human references using TF-IDF weighted n-gram representations, aiming to capture consensus with human-written descriptions.

UniChart is a chart-specific vision-language model that encodes chart text, data, and visual elements and uses a chart-grounded decoder for chart understanding and reasoning tasks (Masry et al., 2023).

ChartInstruct is an instruction-tuned chart model for diverse chart understanding and reasoning tasks (Masry et al., 2024).

Qwen2.5-VL-7B (Team, 2025) is a vision-language model from the Qwen2.5-VL series. It is designed for multimodal understanding tasks, including visual recognition, document parsing, chart understanding, and image-grounded reasoning.

Qwen3-VL-8B (Team, 2025) is a vision-language model from the Qwen3-VL series. It supports interleaved multimodal inputs and is designed for image, video, and text understanding, making it suitable for figure-based evaluation.

Gemini-2.5-Flash (Comanici et al., 2025) is a proprietary multimodal model from the Gemini family. We use it as a strong reference-free evaluator because it supports image-text inputs and is optimized for efficient multimodal reasoning.

Gemini-3.0-Pro (Comanici et al., 2025) is a stronger proprietary multimodal model from the

Subset	Annotation	Samples	Score Label	Usage
Human Seed Training	Human	1,000	5,000	EvaModel and RefModel training
Synthetic Training	Model-generated	9,000	45,000	Training data expansion
HV Golden Test	Human	1,000	5,000	Quality Assessment & Benchmark
Total	–	11,000	55,000	–

Table 5: Data splits of FigGER. HV denotes human-verified.

Gemini family. We include it as a high-capacity reference-free evaluator to test whether general-purpose advanced MLLMs can align with human judgments on fine-grained figure summary evaluation.

GPT-5-Mini (Singh et al., 2025) is a lightweight proprietary model from the GPT series. We include it to evaluate the performance of a compact general model under the same reference-free figure-summary scoring setting.

MiniCPM-V2.6 (Yao et al., 2024) is an efficient multimodal large language model designed for strong visual-language understanding with relatively low deployment cost. We use it as an open-source reference-free baseline.

InternVL-2.5-8B (Chen et al., 2024) is an open-source vision-language model based on the InternVL series. It progressively aligns a large-scale vision foundation model with a language model and is used as another open-source reference-free evaluator.

B.2 Evaluation Metrics

We evaluate all methods on the FigGER human-verified golden test set using four metrics: Pearson Correlation (PC), Spearman Correlation (SC), Mean Absolute Error (MAE), and Mean Squared Error (MSE). For each sample, the predicted score and the ground-truth human score can refer either to a single evaluation dimension or to the overall score aggregated across dimensions.

Let $\hat{y}_i$ denote the predicted score for the i -th sample, and let y_i denote the corresponding ground-truth human score. The total number of evaluated samples is denoted as N . The mean values of predicted and ground-truth scores are denoted as $\bar{\hat{y}}$ and $\bar{y}$, respectively:

$$\bar{\hat{y}} = \frac{1}{N} \sum_{i=1}^N \hat{y}_i, \quad \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i. \quad (11)$$

Pearson Correlation (PC). Pearson Correlation measures the linear correlation between predicted

scores and ground-truth human scores. It is defined as:

$$PC = \frac{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}. \quad (12)$$

A higher PC indicates that the predicted scores have stronger linear consistency with human scores. In our experiments, PC is used to measure whether an evaluation method can assign scores that vary consistently with human judgments.

Spearman Correlation (SC). SC measures the rank-order consistency between predicted scores and ground-truth scores. It is computed by first converting the predicted scores and human scores into ranks. Let $R(\hat{y}_i)$ denote the rank of $\hat{y}_i$ among all predicted scores, and let $R(y_i)$ denote the rank of y_i among all ground-truth scores. SC is defined as:

$$SC = \frac{\sum_{i=1}^N (R(\hat{y}_i) - \overline{R(\hat{y})})(R(y_i) - \overline{R(y)})}{\sqrt{\sum_{i=1}^N (R(\hat{y}_i) - \overline{R(\hat{y})})^2} \sqrt{\sum_{i=1}^N (R(y_i) - \overline{R(y)})^2}} \quad (13)$$

Here, $\overline{R(\hat{y})}$ and $\overline{R(y)}$ denote the average ranks of predicted scores and ground-truth scores, respectively. A higher SC means that the method better preserves the relative ordering of summary quality according to human judgments.

Mean Absolute Error (MAE). MAE measures the average absolute difference between predicted scores and ground-truth scores. It is defined as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|. \quad (14)$$

A lower MAE indicates that the predicted scores are closer to human scores on average. Compared with correlation-based metrics, MAE directly reflects the numerical prediction error.

Mean Squared Error (MSE). MSE measures the average squared difference between predicted

EVALUATE INSTRUCTION

You are a professional expert in figure-summary evaluation, skilled at conducting strict five-dimension evaluations based on multimodal input (image + text).

Based on the input "figure image" and "original summary," you must evaluate the summary along the following five dimensions with a strict 0/1/2 scoring scheme, and provide reasons.

[Scoring Dimension Definitions]

1. Faithfulness:

Ensure the summary strictly adheres to the figure's facts, including values, trends, proportions, and labels; no fabrication, distortion, or misinterpretation.

- 0: The summary severely deviates from the figure's facts, with errors or fabricated content.
- 1: Largely consistent with the figure, but with partial deviations or inaccurate details.
- 2: Fully faithful to the figure's facts, with all information correct.

2. Completeness:

Check whether the summary covers core elements, main trends, and key comparisons in the figure.

- 0: Misses major content or key information.
- 1: Covers the main information but with secondary omissions.
- 2: Fully covers all key information.

3. Conciseness:

Check whether the summary is compact, information-dense, and free of redundancy.

- 0: Redundant, or overly brief to the point of missing key information.
- 1: Generally concise but can be further optimized.
- 2: Tersely written with efficient expression.

4. Logicality:

Check whether the logic is coherent, sequence is reasonable, and causal relations are clear.

- 0: Logical confusion or self-contradiction.
- 1: Basically coherent but with slight jumps or ambiguity.
- 2: Clear logic and strong consistency.

5. Analysis:

Check whether the summary meets domain-standard expression (e.g., finance, statistics, economics).

- 0: Lacks professionalism, or uses incorrect terminology/units.
- 1: Professional expression is insufficient or slightly colloquial.
- 2: Accurate terminology, formal tone, and analytical depth.

[Original Summary]

{summary}

[Output Requirements]

1. You must output standardized JSON inside ``<evaluation>`` tags, with the following structure:

```
<evaluation>
{"scores": {"faithfulness": 0-2, "completeness": 0-2, "conciseness": 0-2, "logicality": 0-2, "analysis": 0-2},
"reasons": {"faithfulness": "brief explanation", "completeness": "brief explanation", "conciseness": "brief explanation", "logicality": "brief explanation", "analysis": "brief explanation"}}
</evaluation>
```

2. The JSON must be valid, single-line JSON.

3. The output format must include ``<evaluation>`` tags containing scores, reasons, and weights.

Figure 8: Evaluation instruction used for reference-free model evaluation.

scores and ground-truth scores. It is defined as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2. \quad (15)$$

A lower MSE indicates smaller prediction errors. Since larger errors are squared, MSE penalizes

large deviations more strongly than MAE.

B.3 Evaluation Instruction

The detailed evaluation instruction used in our experiments is shown in Figure 8.